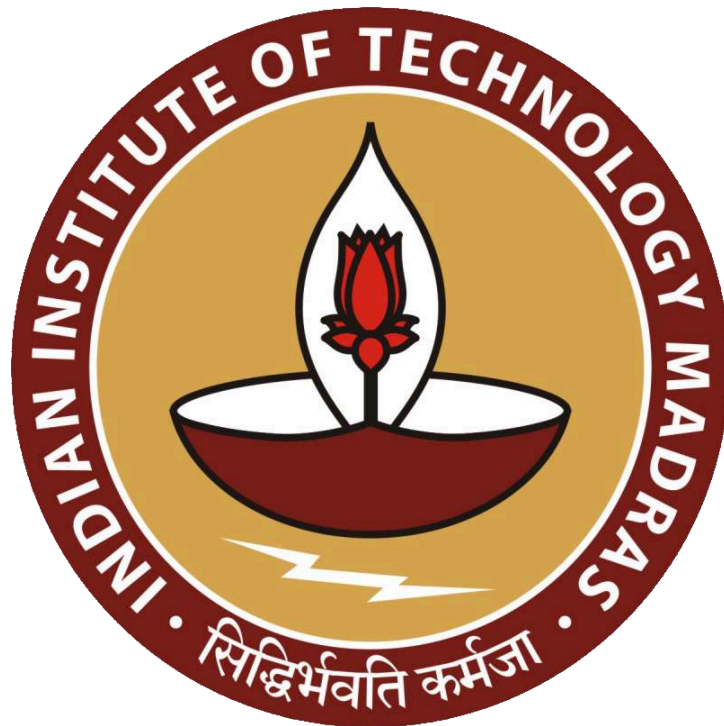


Optimizing Retail Efficiency: A Data-Driven Approach for
LAXMI GENERAL STORE

A Final Term Report for the BDM Capstone Project

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1.Executive Summary and Title

Laxmi General Store, a small grocery outlet, faces several operational challenges, primarily revolving around inefficient inventory management, unclear customer spending patterns, frequent stockouts of popular items, and suboptimal budget allocation. This report analyzes the store's sales data from March to May 2024 to address these concerns and provide data-driven solutions.

Key findings show that essential products such as coffee, chocolates, sugar, personal care and toiletries account for a significant portion of the store's total sales. The report proposes utilizing **Reorder Points** to prevent stockouts of these high-demand items and maintain adequate inventory levels. Additionally, insights from the data on customer spending patterns can help optimize the store's budget allocation, focusing on fast-moving goods to reduce operational costs and increase customer satisfaction.

By adopting these data-driven strategies, the store can minimize holding and ordering costs, improve stock availability, and reduce the risk of lost sales due to out-of-stock items. These improvements, along with an optimized budget allocation model, will enhance overall profitability and operational efficiency.

The recommendations include reallocating the budget towards high-demand products, forecasting seasonal demand to better manage stock levels, and tracking daily sales trends for proactive inventory management. These steps will ensure a more efficient operation, increased customer satisfaction, and improved financial performance for Laxmi General Store.

Datasets

2.Detailed Explanation of Analysis Process/Method

Data Overview

The datasets provided contained the following fields for each month (March, April, and May 2024):

Date: The transaction date.

Item Name: The name of the purchased product.

Quantity: The number of units purchased.

Unit Price: Price per unit of each item.

Amount: Total spending on that item (calculated as Quantity × Unit Price).

Monthly Sales Analysis

The first step in analyzing the store's sales was calculating total sales for each month. Using the sales data for March, April, and May 2024, the total revenue was calculated using Excel's **SUM** formula. The formula used was:

$$\text{Total Sales} = \Sigma(\text{Amount})$$

In March 2024, total sales reached ₹9,18,193, followed by a slight dip in April to ₹9,12,966. However, sales saw a significant rise in May, climbing to ₹10,14,264, largely driven by successful promotional activities, including discounts on selected items offered by the owner of Laxmi General Store. This trend highlights the need for the store to anticipate periodic sales peaks and ensure adequate inventory levels during high-demand periods such as May.

Top - Selling Products

To identify which products contributed most to overall sales, Excel's **SORT** and **FILTER** functions were applied to the dataset. This helped highlight the top-selling items based on the total revenue generated for each product. Items such as **coffee, chocolates, sugar, personal care and toiletries** consistently ranked among the highest sellers across all three months.

Using the **SUMIF** formula in Excel, we further calculated total sales for each top-selling product over the three months. The formula used was:

Total Product Sales=SUMIF(Item Name,Product,Amount)

This calculation confirmed that groceries, particularly **coffee, chocolates, sugar, personal care, rice, flour and toiletries are essential** drivers of revenue. The high sales figures for these products emphasize their critical role in the store's profitability and highlight the importance of ensuring consistent stock availability.

Average Transaction Value

The next step was to determine the average amount spent by customers per transaction. This metric provides insight into typical customer behavior and helps gauge whether most customers are making large or small purchases. The average transaction value was calculated using the following formula:

Average Transaction Value=Number of Transactions/Total Sales

For March, with total sales of ₹9,18,193 spread across 1,765 transactions, the average transaction value was ₹520. In April, the value was slightly lower at ₹515, and in May, it rose to ₹540. This steady range (₹500–₹540) indicates that customers are making consistent purchases in terms of value, suggesting stable purchasing behavior over time.

Stockout Prevention for Popular Items

To address the issue of frequent stockouts, particularly for high-demand products like rice,

flour and toiletries the data was examined to determine how often these items were purchased. Excel's **COUNTIF** function was used to count the number of times each product appeared in the dataset. For instance, rice appeared in 24 transactions in March, 22 in April, and 26 in May. The corresponding sales volume for each month was calculated using the **SUMIF** formula, as mentioned earlier.

The high sales frequency and volume for products like rice and toiletries suggest that these items are consistently in demand and must be restocked regularly to avoid stockouts. Based on this analysis, the store should implement a predictive restocking system that tracks daily sales trends to estimate future demand. By forecasting when stock levels will fall below a certain threshold, the store can place timely replenishment orders to prevent shortages. A possible formula for tracking this in Excel could be:

$$\text{Reorder Point} = \text{Average Daily Sales} \times \text{Lead Time} + \text{Safety Stock}$$

Where:

- **Average Daily Sales** is derived from past sales data.
- **Lead Time** is 7 days as mentioned by the owner.
- **Safety Stock** is a buffer to account for unexpected demand spikes (not available)
- **The average daily sales of rice are 2 units and the lead time is 7 days, the store should reorder when stock falls below 14 units, plus any safety stock.**

Budget Allocation for Inventory

The next part of the analysis focused on how the store's budget should be allocated across product categories (categorized using python) to prevent both overstocking and understocking. Excel's **SUMIFS** function was used to calculate total spending by category (groceries, toiletries, cleaning supplies) for each month:

$$\text{Total Category Spending} = \text{SUMIFS}(\text{Amount}, \text{Category}, \text{Category Name})$$

For instance, in March, total spending on groceries amounted to ₹1,68,434, while spending on toiletries was ₹9,003. This comparison shows that groceries form the largest proportion of the store's revenue, followed by toiletries and cleaning supplies.

Given that groceries account for the bulk of sales, a larger portion of the store's budget should be allocated to purchasing these items. Conversely, items in categories with lower demand (such as luxury personal care or specialized cleaning products) should receive less budgetary focus to avoid overstocking. Budget allocation can be better managed by comparing the monthly sales trend of each category against the budget currently assigned to it. This will allow the store to adjust spending in line with demand, ensuring that popular products are well-stocked and that capital is not wasted on slower-moving inventory.

Seasonal Demand and Trend Analysis

A deeper analysis of the sales data also revealed seasonal demand fluctuations, particularly for cleaning supplies, which saw a spike in May with total sales of ₹1,330. This suggests that certain products (such as detergents, Harpic and Dettol handwash) experience periodic demand surges, potentially due to discounts. To further explore this, a line graph was created in Excel, mapping the total monthly sales for each product category. This visual representation made it clear that cleaning supplies and certain toiletries exhibited seasonal demand patterns, with increased sales in specific months.

By understanding these trends, the store can better align its inventory and budget planning with customer behavior. For example, in preparation for seasonal demand spikes in cleaning supplies, the store can increase its stock in the preceding month, ensuring that products are available when customer demand peaks.

Dairy Products Spending Comparison per Month

I then analyzed the spending on dairy products across the months of March, April, and May (as it holds a pretty good proportion in shop's sales) demonstrates a relatively stable pattern, though there are subtle variations that can be attributed to different factors such as

seasonal consumption or the availability of specific products. In March, the total expenditure on dairy products amounted to ₹34,468. This is consistent with regular household consumption patterns, as dairy products are considered essential in most households. The range of products likely included milk, cheese, and dairy whitener, all of which are staples in the diet of many families. The steady expenditure during this month suggests that there was no significant disruption in purchasing habits, and consumers maintained their usual level of dairy consumption.

In April, the spending on dairy products slightly decreased to ₹32,697. This dip may reflect changes in consumer behavior during the month, possibly due to a shift in priorities or availability. The decline is minimal, but it indicates that there could have been a slight reduction in the need for certain dairy items, possibly due to household stockpiling in previous months or alternative food consumption patterns emerging in this period. April also marks a transitional phase in the season, which could have influenced purchasing decisions, with consumers perhaps opting for lighter alternatives as temperatures rise.

By May, spending on dairy products decreased to ₹32,079, which is slightly lower than March. This decline could be attributed to reduced demand for dairy-related items as summer intensifies. Although hot weather typically drives up consumption of cold dairy products, such as curd, yogurt, and ice cream, other factors like shifting consumer preferences or cost considerations might have led to the dip in expenditure. Overall, the steady spending on dairy across the months underscores its essential role in household purchases, while the variations provide insights into how external factors, such as seasonality, may influence consumer behavior.

Weekly Sales Trends

To gain a more detailed understanding of sales fluctuations, the data was grouped by week using Excel's **WEEKNUM** function. Weekly sales totals were then calculated, revealing significant variations in sales across the different weeks.

The analysis showed that certain weeks had noticeably higher sales, such as weeks 10, 14, and 16, with lower sales in weeks 9 and 17. This pattern suggests that customer demand

peaks during specific weeks, possibly due to external factors like payday cycles or promotional events. The **SUMIFS** function was used to calculate weekly sales totals, ensuring accurate weekly breakdowns.

By understanding these weekly sales trends, the store can better forecast demand and ensure that stock levels are adjusted accordingly. For instance, if cleaning supplies and toiletries consistently see higher demand during the second week of each month, the store should plan its restocking schedule to ensure sufficient stock is available before this surge.

Below are the key observations

1. Popular Items Across Months

In March, products like *AMUL Cheese Block 200g* and *Tide Jasmine & Rose Detergent Powder* were frequently purchased, showing a mix of dairy products and household cleaning items. In April, *BB Jim Jam* biscuits and *Parle 20 20 Coconut Biscuits* were among the top purchases, indicating a preference for snack items. In May, products like *Amulya Dairy Whitener* and *Haldiram's Rasgulla* were common, showing a shift towards beverages and sweets.

2. Price Variations and Unit Price Trends

March had notable higher-cost items, such as *Fresh Cashew Nut 250g* at ₹365, which is significantly more expensive than the average daily products like biscuits or detergent. April data shows consistent pricing of snacks and household goods, with most items falling under ₹100, except for larger packs such as the *Parle Coconut Biscuits*. May had an interesting observation of the unit price of smaller items like *Amulya Dairy Whitener* priced at ₹5 per unit, indicating a preference for affordable single-use products.

3. Purchasing Quantity and Patterns

High-quantity purchases are visible, especially for smaller or frequently used items. For

example, in March, 5 units of *BB Marie Gold* were bought at once, suggesting stocking up on staples. In May, large quantities were also observed for single-use items like *Amulya Dairy Whitener* (23 units), which might indicate bulk purchasing for regular use. Across all months, we see household cleaning and staple food items consistently being purchased in small to moderate quantities, indicating regular household shopping habits.

4. Spending Patterns Across Categories

The highest-spending items appear to be concentrated in food categories, with products like cashew nuts, sweets (e.g., *Haldiram's Rasgulla*), and large packs of biscuits. The spending on snacks and specialty food items appears significant. Cleaning and personal care items maintain a steady flow of moderate spending, highlighting their necessity but lower price per unit compared to specialty foods.

5. Seasonal Purchase Observations

There is a noticeable shift from staple foods and cleaning products in March to more snack and sweet products in May, which could be reflective of festival preparations or changing consumer preferences as summer sets in.

6. The comparison between weekday and weekend sales reveals the following:

1. Average sales:

- a. Weekdays: ₹32,193.79
- b. Weekends: ₹31,524.35

2. Sales range:

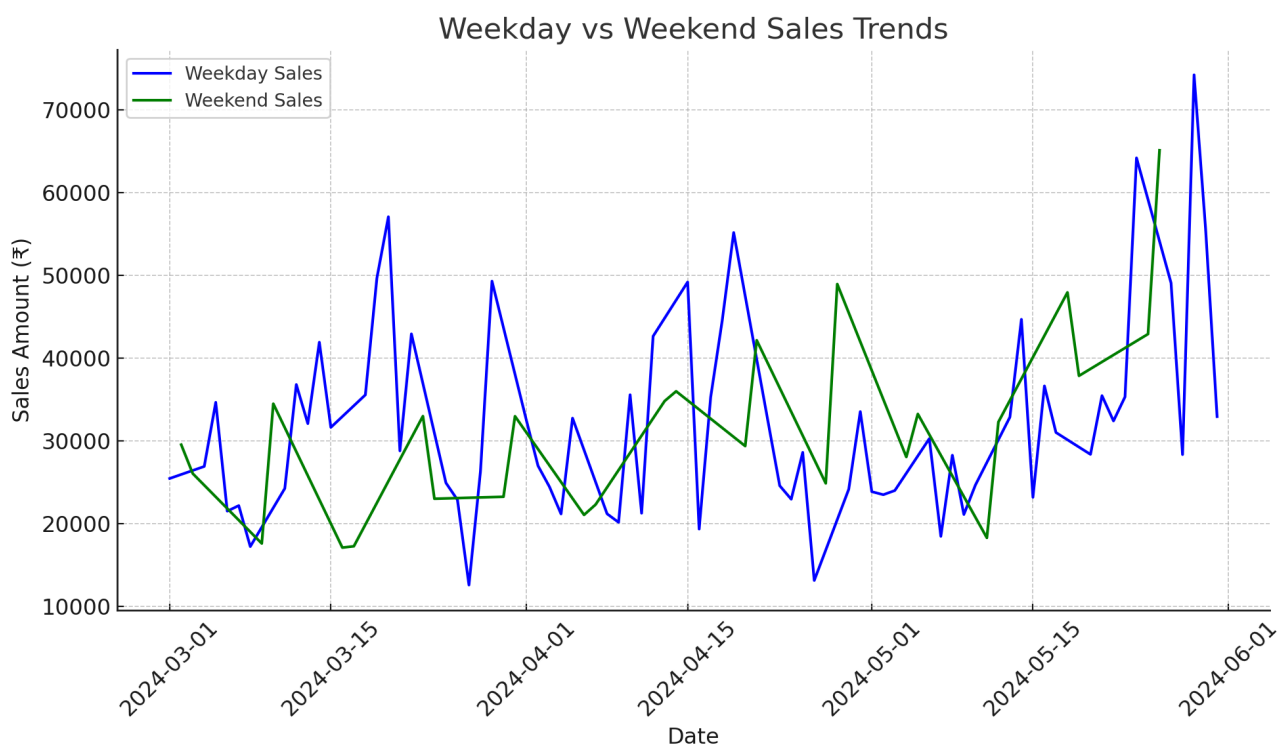
- a. Weekdays: ₹12,606 (min) to ₹74,192 (max)
- b. Weekends: ₹17,123 (min) to ₹65,090.50 (max)

3. Sales variability:

- a. Weekdays have a slightly higher standard deviation (₹12,135) compared to weekends (₹11,377), indicating more fluctuation in weekday sales.

While weekday sales have a marginally higher average, the overall trends are similar, with both weekdays and weekends showing significant variability in sales amounts.

3. Results and Findings (Graphs and other Pictorial Representation)



1. General Trend:

Both weekday and weekend sales fluctuate, but there is a noticeable increasing trend in sales from early March to early June, with some sharp peaks, especially in weekday sales.

2. Weekday vs Weekend Sales:

Weekday sales (blue line) tend to have more volatility, showing higher peaks and dips compared to weekend sales (green line). Weekend sales are relatively more stable, showing less frequent sharp changes, but also steadily increasing in the later months.

3. March (2024-03-01 to 2024-03-31):

Weekday sales start lower, with a significant upward trend after mid-March.

However, weekends perform similarly or slightly better in the first half of March, as weekend sales seem to be higher than weekday sales until mid-March.

4. April (2024-04-01 to 2024-04-30):

Weekday sales have some of the highest peaks, especially around mid-April, where sales reach the highest point of the graph, while weekend sales remain more moderate. Weekends seem to lag behind weekday sales in April, indicating stronger weekday performance.

5. May (2024-05-01 to 2024-05-31):

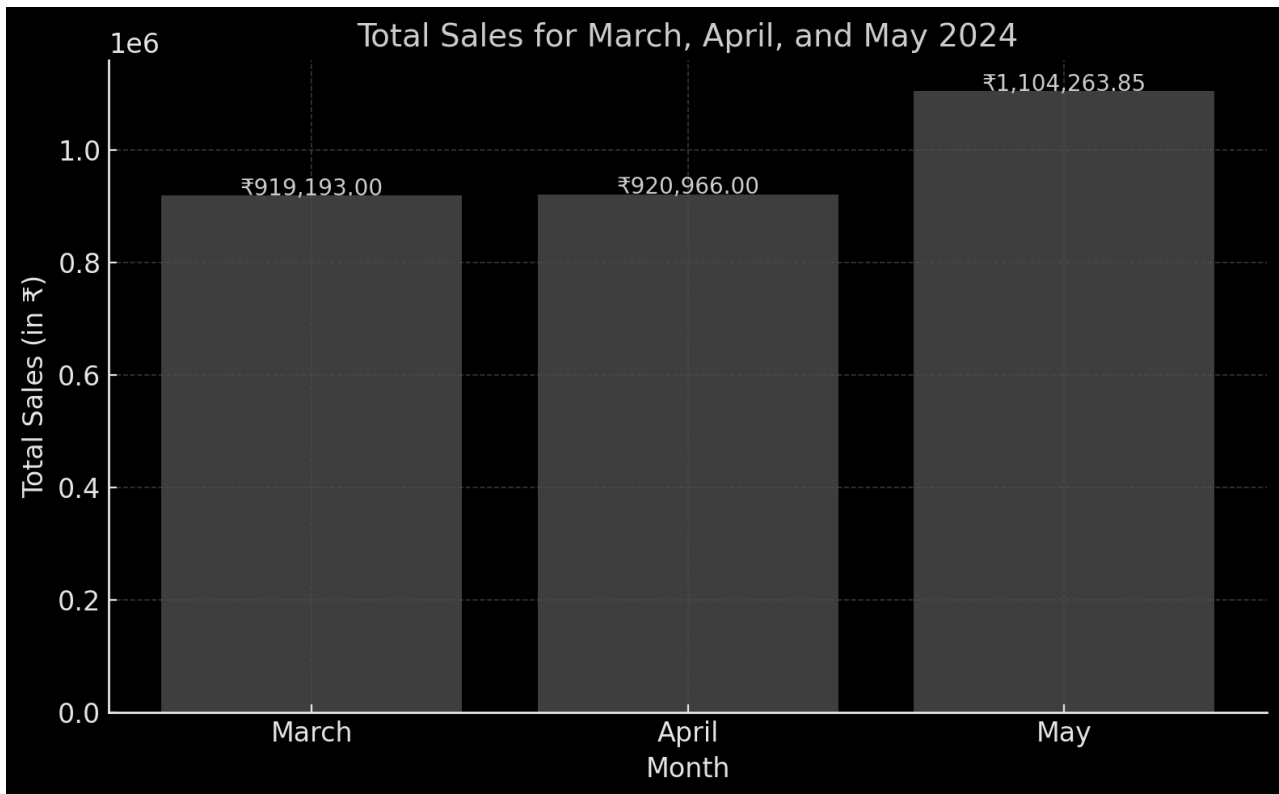
May shows a more intertwined pattern, where weekend and weekday sales intersect frequently. Toward the end of May, weekend sales seem to outperform weekdays in certain periods. Weekdays still experience some peaks, but not as sharp or frequent as in previous months.

Key Observations:

Weekday sales are generally more volatile with higher peaks, especially in mid-April and June.

Weekend sales are more stable, with less frequent spikes but catch up toward the end of the observed period.

April shows a strong performance in weekday sales, while May shows a more balanced trend between weekdays and weekends.

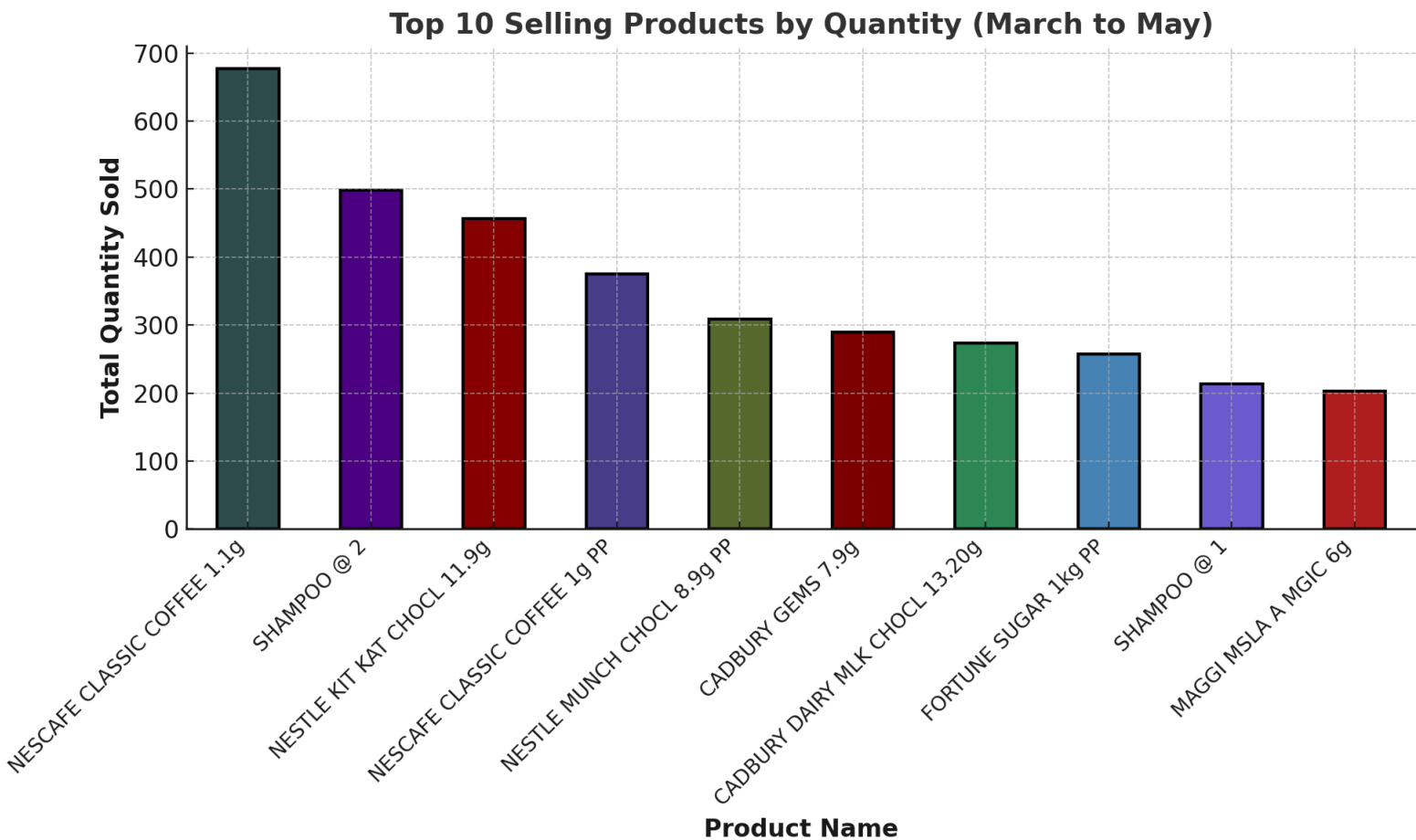


Graph 2: Bar Graph showing Total Sales for March, April and May

The graph showing total sales for March, April, and May 2024 reveals several important insights:

1. **May as the Best-Performing Month:** May recorded the highest total sales among the three months. This increase could be due to higher customer demand, promotional campaigns, or product launches during this period. The sharp rise in sales suggests that May was a successful month for the business, likely benefiting from strategic marketing or seasonal trends.
2. **April's Sales Decline:** April had the lowest total sales. This dip could result from reduced marketing efforts, fewer customer engagements, or external factors like holidays or reduced consumer spending. It's worth investigating why sales dropped in April to avoid similar dips in the future.
3. **Steady Growth from March to May:** There is a clear upward trend from March to May, indicating overall business growth during this period. March served as a baseline for steady sales, while May showed significant improvement. This

positive growth trend suggests that the business has the potential to continue building momentum and expanding its market presence.



Graph 3: Bar Graph showing Top 10 Selling Products by Quantity

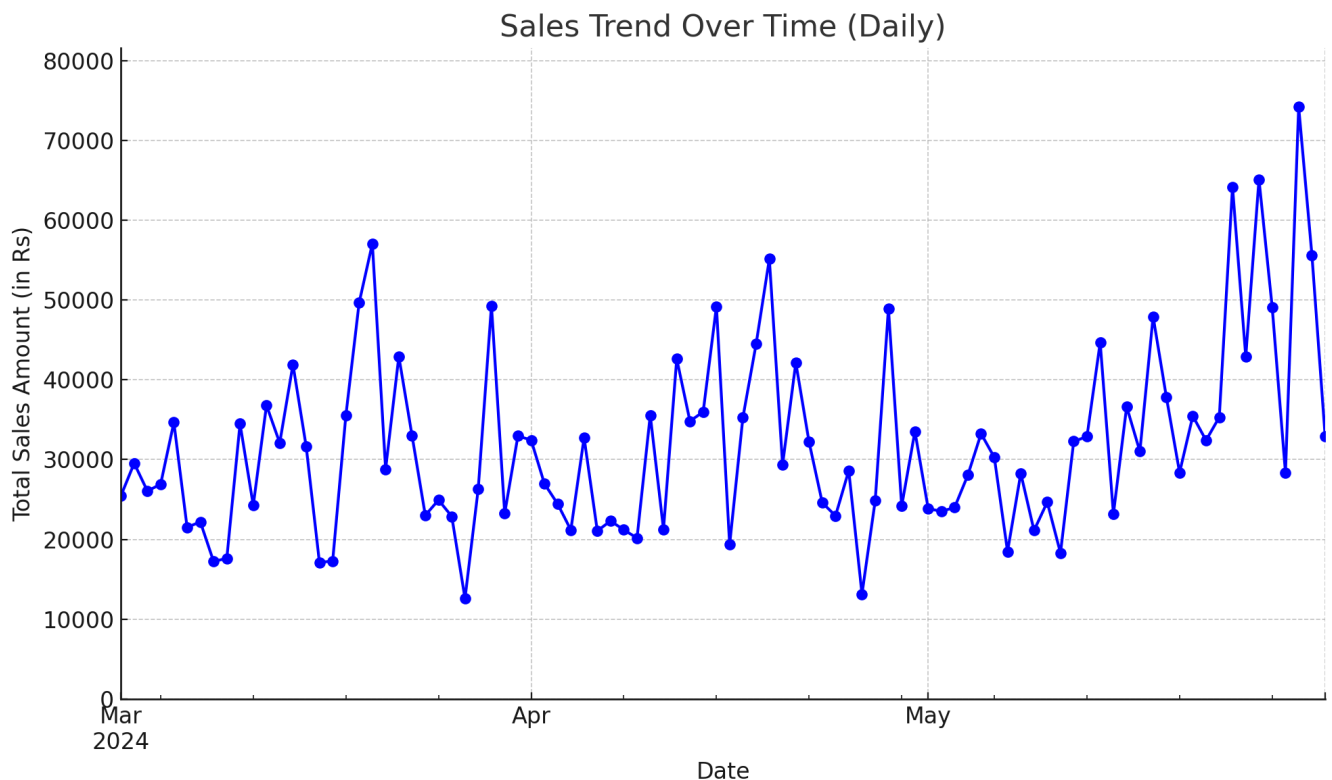
The bar chart illustrates the top 10 selling products by quantity from March to May. Here are some key inferences:

1. **NESCAFÉ CLASSIC COFFEE 1.1g** is the top-selling product with over 700 units sold, significantly outperforming the others.
2. **SHAMPOO 8** ranks second with around 500 units sold, showing strong sales in a non-food category.
3. **NESTLE KIT KAT CHOCOLATE 11.8g** and **NESCAFÉ CLASSIC COFFEE 18.9g**

also performed well, both with over 400 units sold.

4. Other products such as **CADBURY DAIRY MILK CHOCOLATE**, **FORTUNE SUGAR**, and **MAGGI MASALA** have lower but consistent sales, each ranging between 250 and 350 units.
5. The chart demonstrates a mixture of product categories (coffee, chocolate, shampoo, and sugar), indicating a diverse range of popular items during this period.
6. There's a clear dominance of **NESCAFÉ** products, which appear multiple times in the top 10 list, highlighting strong consumer demand for coffee.
7. The distribution of quantities sold shows that while the leading products have significant sales volumes, the difference between the lower-ranked products is relatively marginal.

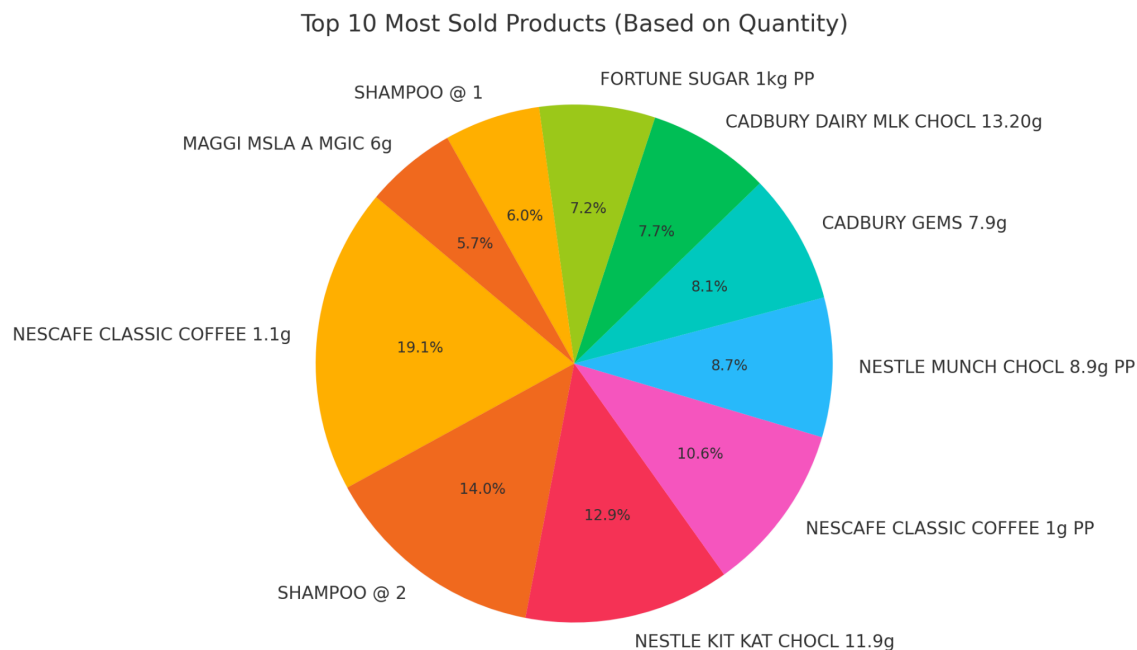
This suggests that **NESCAFÉ products** and **personal care items** like shampoo are particularly strong sellers during the period analyzed.



Graph 4: Line Graph showing Sales Trend Over Time (Daily)

The above graph depicts a daily sales trend over the period from March 2024 to May 2024, showing the total sales amount in Indian Rupees (Rs). Key observations include:

1. **Fluctuating Sales:** The sales exhibit high volatility, with numerous peaks and troughs throughout the period.
2. **Increasing Trend:** Despite the fluctuations, there is a noticeable upward trend in sales, especially from mid-April onwards.
3. **Peaks in Sales:** Several spikes in sales can be seen, particularly towards the end of the graph in May, where sales reach around 70,000 Rs.
4. **Low Points:** There are also sharp declines in sales at various points, with some days showing lower sales around 10,000-15,000 Rs.



Graph 5: Pie Chart showing budget allocation based on sales data

1. Groceries

Groceries are essential items that consumers purchase regularly, forming the backbone of sales in any store. The top products in this category include **GMC Sugar 1kg**, **Everest Garam Masala 50g**, **Bregano Sooji 500g**, and **Fresh Cashew Nuts 250g**. These are basic pantry staples, including sweeteners, spices, grains, and nuts. Given

their constant demand and daily usage in households, groceries should command a substantial portion of the budget, approximately **30-35%**. Ensuring a steady supply of these items is critical to meeting customer needs, and stocking them well will prevent potential revenue loss from out-of-stock situations.

2. Snacks

Snacks are another significant category, driven by frequent but small purchases. This includes items like **Amul Cheese Block 200g**, **BB Marie Gold 73g**, **Sunfeast Mom's Magic Cashew 197g**, and **Cadbury 5Star Chocolate 24g**. These products are generally consumed quickly and are often impulse buys. They provide variety in a shopping cart but don't typically require a large investment compared to groceries. However, due to their popularity, allocating around **20-25%** of the budget to snacks will ensure that you have sufficient stock to cater to the frequent demand, while still balancing it with other essential categories.

3. Toiletries

Toiletries, including **Tide + Double Power Jasmine & Rose Detergent Powder 1kg** and **Godrej Ezee Liquid Detergent 250g**, are household necessities used for cleaning and laundry. While these items are essential, they are purchased less frequently than food products. Therefore, the budget for toiletries should be moderate, around **15-20%**. Ensuring that the stock levels for these products remain stable is important, but since they are not daily consumables, they don't need as large a share as groceries or snacks.

4. Dairy

Dairy-related products, such as **Amulya Dairy Whitener 10g**, cater to a niche group of consumers, particularly those who prefer a milk substitute in their beverages. Although dairy products are consumed daily, the dairy whitener in this case is a more specialized product, likely bought by a smaller portion of customers. Therefore, allocating **10-15%** of the budget to dairy should suffice, ensuring that you cater to the specific needs of these customers while not overcommitting resources to a smaller market.

5. Confectionery

Finally, confectionery products like **Haldiram's Rasgulla 1.25kg** fall under occasional purchases, typically bought during celebrations or as indulgent treats. These products do not have the same frequency of demand as other categories. Hence, it is advisable to allocate a smaller portion of the budget, around **5-10%**, for confectionery items. Stocking these products is important, particularly during festive seasons, but they should not take priority over more frequently consumed categories.

Budget Allocation Strategy

To optimize your budget, focus on allocating the largest portion to essential items like

groceries and snacks. These categories ensure constant traffic and repeat purchases. Toiletries and dairy should receive moderate allocation due to their less frequent but still essential nature. Confectionery, being an occasional purchase, requires the least budget but should not be neglected, especially during peak sales periods like holidays. Adjusting your budget seasonally and based on historical demand can help in better managing stock levels and sales performance.

4. Interpretation of Results and Recommendation

Problem 1: Unclear Spending Patterns

Results:

1. **Graph 3: Bar Graph of Top 10 Selling Products by Quantity:** Shows that **NESCAFÉ CLASSIC COFFEE 1.1g** is the top-selling product, followed by **SHAMPOO 8** and **NESTLE KIT KAT CHOCOLATE**. This suggests that customers frequently purchase coffee, personal care, and snack items.
2. **Graph 4: Line Graph Showing Daily Sales Trends:** This shows a fluctuating but overall increasing trend in daily sales from March to May, indicating periods of high and low spending without clear categorization trends.

Recommendations:

1. **Customer Segmentation:** Based on sales data, categorize products into high, medium, and low demand. For example, **NESCAFÉ products** and **shampoos** should be classified as high-demand items. This will help in identifying which categories (groceries, personal care, snacks, etc.) drive the majority of sales and focus on stocking and promoting these products.
2. **Personal Care Focus:** As personal care items like shampoo are among the top sellers, allocate more budget to such products, while also tracking seasonal variations (Graph 4). This will help in optimizing the inventory and improving profitability.

Problem 2: Stockout Woes

Results:

1. **Graph 2: Bar Graph of Total Sales for March, April, and May:** May shows the highest sales, and given the demand spikes in May, frequent stockouts for popular items such as rice and toiletries are likely.
2. **Graph 3** further highlights the importance of personal care and grocery items as popular purchases, which could be prone to stockouts.

Recommendations:

1. **Predictive Stocking Using Reorder Points:** As suggested in the report, use the **Reorder Point = Average Daily Sales x Lead Time + Safety Stock** formula. For example, rice's average daily sales is 2 units with a lead time of 7 days, meaning you should reorder rice when stock falls below 14 units.

2. **Daily Tracking of High-Demand Items:** Implement regular tracking of high-selling products like **rice**, **flour**, and **toiletries** to avoid stockouts. Data shows a consistent demand for these items, so ensuring stock availability is crucial for maintaining customer satisfaction and preventing loss of sales.

Problem 3: Budgeting Blindspot

Results:

1. **Graph 5: Pie Chart of Budget Allocation:** Reveals the budget distribution with groceries receiving the largest portion (30-35%), followed by snacks (20-25%), toiletries (15-20%), dairy (10-15%), and confectionery (5-10%).
2. **Graph 3** and **Graph 5** show that groceries and personal care products are driving sales, justifying their significant share of the budget.

Recommendations:

1. **Dynamic Budget Allocation:** Adjust the store's budget allocation dynamically by aligning it with sales trends from **Graph 5**. Allocate more funds toward high-performing categories (e.g., groceries and toiletries) and reduce spending on low-demand items such as confectionery.
2. **Seasonal Budget Adjustments:** As indicated by **Graph 4**, sales tend to spike in specific periods (e.g., May), so increase the budget for high-demand items during these months. This ensures that you are capitalizing on increased customer demand while avoiding overspending during low-demand periods.

Conclusion

To address the Laxmi General store's key challenges:

1. **Clear Categorization of Spending Patterns** (via top-sellers analysis and daily sales trends).
2. **Prevent Stockouts** (with predictive stock tracking and reorder points for high-demand products).
3. **Effective Budgeting** (with dynamic allocations based on historical sales data and seasonal trends).